



Laboratory work №3

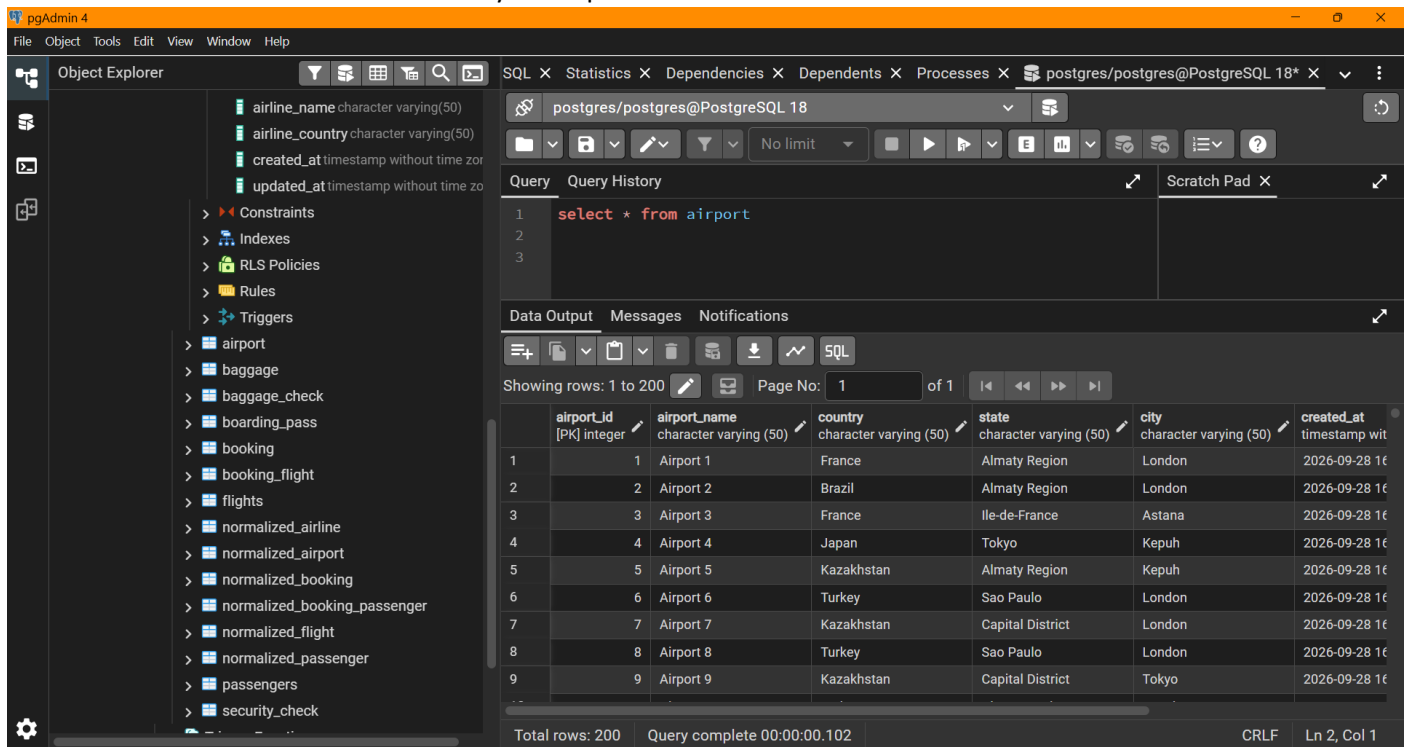
Databases

Tolepказы Galamat

Tuesday, 11:00

Almaty 2026

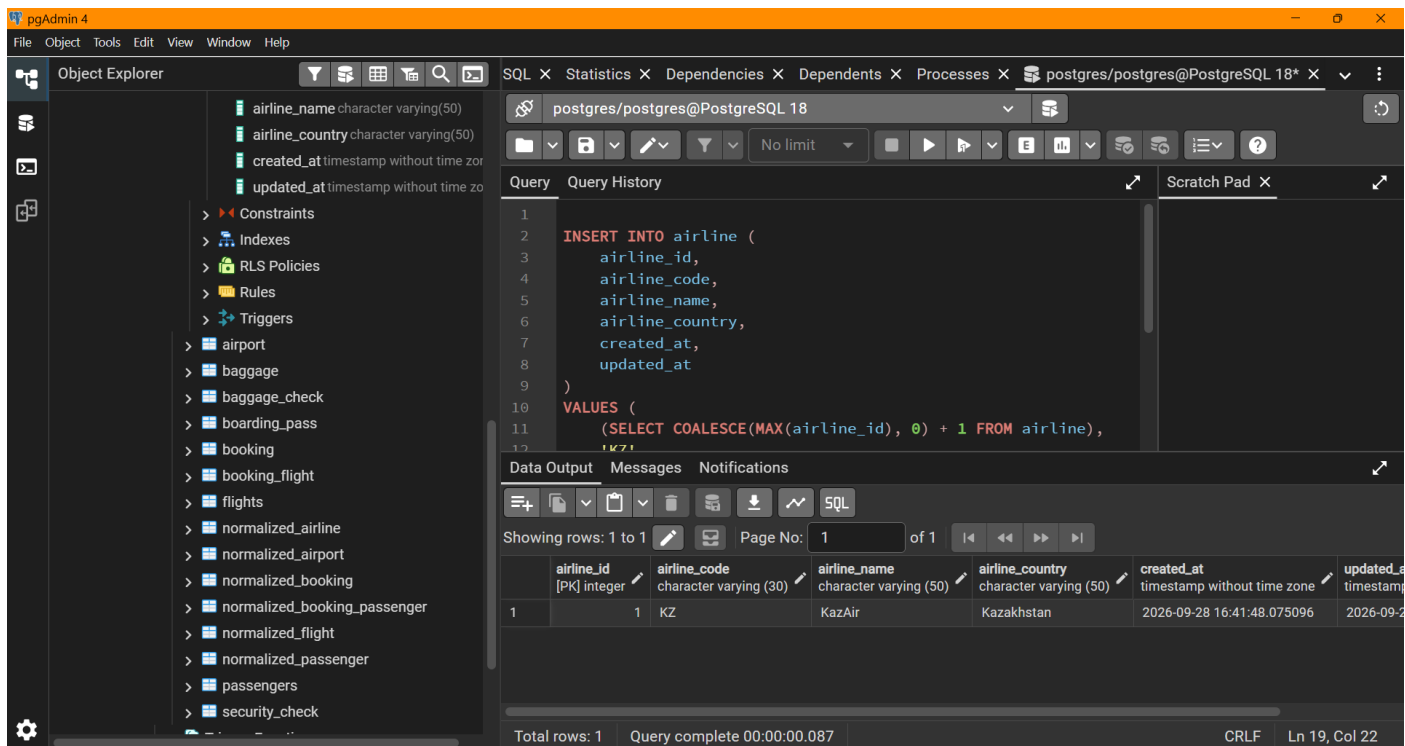
1. Generate and insert 200 random rows in your airport database



The screenshot shows the pgAdmin 4 interface. The left pane displays the 'Object Explorer' with the 'airport' table selected. The right pane shows the 'Query' editor with the SQL statement `select * from airport`. The 'Data Output' pane displays the first 9 rows of the 'airport' table. The status bar at the bottom indicates 'Total rows: 200' and 'Query complete 00:00:00.102'.

airport_id	airport_name	country	state	city	created_at
1	Airport 1	France	Almaty Region	London	2026-09-28 16:41:48.075096
2	Airport 2	Brazil	Almaty Region	London	2026-09-28 16:41:48.075096
3	Airport 3	France	Ile-de-France	Astana	2026-09-28 16:41:48.075096
4	Airport 4	Japan	Tokyo	Kepuh	2026-09-28 16:41:48.075096
5	Airport 5	Kazakhstan	Almaty Region	Kepuh	2026-09-28 16:41:48.075096
6	Airport 6	Turkey	Sao Paulo	London	2026-09-28 16:41:48.075096
7	Airport 7	Kazakhstan	Capital District	London	2026-09-28 16:41:48.075096
8	Airport 8	Turkey	Sao Paulo	London	2026-09-28 16:41:48.075096
9	Airport 9	Kazakhstan	Capital District	Tokyo	2026-09-28 16:41:48.075096

2. Add a new airline named "KazAir" based in "Kazakhstan" to the airline table



The screenshot shows the pgAdmin 4 interface. The left pane displays the 'Object Explorer' with the 'airline' table selected. The right pane shows the 'Query' editor with the SQL statement `INSERT INTO airline ( airline_id, airline_code, airline_name, airline_country, created_at, updated_at ) VALUES ( (SELECT COALESCE(MAX(airline_id), 0) + 1 FROM airline), 'KZ', 'KazAir', 'Kazakhstan', '2026-09-28 16:41:48.075096', '2026-09-28 16:41:48.075096' )`. The 'Data Output' pane displays the first row of the 'airline' table. The status bar at the bottom indicates 'Total rows: 1' and 'Query complete 00:00:00.087'.

airline_id	airline_code	airline_name	airline_country	created_at	updated_at
1	KZ	KazAir	Kazakhstan	2026-09-28 16:41:48.075096	2026-09-28 16:41:48.075096

3. Update the airline country "KazAir" to "Turkey".

The screenshot shows the pgAdmin 4 interface. The left pane displays the database schema with tables like airport, baggage, booking, flights, and normalized_* tables. The central SQL editor contains the following query:

```
1
2 UPDATE airline
3 SET airline_country = 'Turkey',
4   updated_at = CURRENT_TIMESTAMP
5 WHERE airline_name = 'KazAir';
6
7 select * from airline
8
9
```

The bottom pane shows the 'Data Output' tab with a single row of data:

airline_id [PK] integer	airline_code character varying (30)	airline_name character varying (50)	airline_country character varying (50)	created_at timestamp without time zone	updated_at timestamp
1	1	KZ	Turkey	2026-09-28 16:41:48.075096	2026-09-28 16:41:48.075096

Total rows: 1 | Query complete 00:00:00.080 | CRLF | Ln 5, Col 31

4. Add three airlines at once: "AirEasy" in "France", "FlyHigh" in "Brazil" and "FlyFly" in "Poland".

The screenshot shows the pgAdmin 4 interface. The left pane displays the database schema. The central SQL editor contains the following query:

```
13
14 airline_code,
15 airline_name,
16 airline_country,
17 CURRENT_TIMESTAMP,
18 CURRENT_TIMESTAMP
19 FROM (
20 VALUES
21 ('AE', 'AirEasy', 'France'),
22 ('FH', 'FlyHigh', 'Brazil'),
23 ('FF', 'FlyFly', 'Poland')
24 ) AS new_airlines (airline_code, airline_name, airline_country);
```

The bottom pane shows the 'Data Output' tab with four rows of data:

airline_id [PK] integer	airline_code character varying (30)	airline_name character varying (50)	airline_country character varying (50)	created_at timestamp without time zone	updated_at timestamp
1	1	KZ	Turkey	2026-09-28 16:41:48.075096	2026-09-28 16:41:48.075096
2	2	AE	France	2026-09-28 16:45:24.458418	2026-09-28 16:45:24.458418
3	3	FF	Poland	2026-09-28 16:45:24.458418	2026-09-28 16:45:24.458418
4	4	FH	Brazil	2026-09-28 16:45:24.458418	2026-09-28 16:45:24.458418

Total rows: 4 | Query complete 00:00:00.088 | CRLF | Ln 23, Col 65

5. Delete all flights whose arrival in 2024 year.

The screenshot shows the pgAdmin 4 interface. On the left, the Object Explorer displays a tree of database objects, including 'airline_name', 'airline_country', 'created_at', 'updated_at', 'Constraints', 'Indexes', 'RLS Policies', 'Rules', 'Triggers', 'airport', 'baggage', 'baggage_check', 'boarding_pass', 'booking', 'booking_flight', 'flights', 'normalized_airline', 'normalized_airport', 'normalized_booking', 'normalized_booking_passenger', 'normalized_flight', 'normalized_passenger', 'passengers', and 'security_check'. The 'flights' table is selected. The main pane shows a SQL query editor with the following query:

```
1  
2 DELETE FROM flights  
3 WHERE act_arrival_time >= TIMESTAMP '2024-01-01 00:00:00'  
4 AND act_arrival_time < TIMESTAMP '2025-01-01 00:00:00';
```

The query is executed, and the Messages pane shows the following message:

```
DELETE 0  
  
Query returned successfully in 61 msec.
```

The status bar at the bottom indicates 'Total rows: Query complete 00:00:00.061' and 'CRLF Ln 4, Col 59'.

6. Increase the price of all tickets in booking table for flights by 15%.

The screenshot shows the pgAdmin 4 interface. On the left, the Object Explorer displays a tree of database objects, including 'airline_name', 'airline_country', 'created_at', 'updated_at', 'Constraints', 'Indexes', 'RLS Policies', 'Rules', 'Triggers', 'airport', 'baggage', 'baggage_check', 'boarding_pass', 'booking', 'booking_flight', 'flights', 'normalized_airline', 'normalized_airport', 'normalized_booking', 'normalized_booking_passenger', 'normalized_flight', 'normalized_passenger', 'passengers', and 'security_check'. The 'booking' table is selected. The main pane shows a SQL query editor with the following query:

```
1 UPDATE booking  
2 SET ticket_price = ticket_price * 1.15,  
3 updated_at = CURRENT_TIMESTAMP;
```

The query is executed, and the Messages pane shows the following message:

```
UPDATE 0  
  
Query returned successfully in 54 msec.
```

The status bar at the bottom indicates 'Total rows: Query complete 00:00:00.054' and 'CRLF Ln 3, Col 36'.

7. Delete all tickets where price is less than 10000.

The screenshot shows the pgAdmin 4 interface. On the left, the Object Explorer displays a tree of database objects, including tables like 'airline_name', 'airline_country', 'created_at', and 'updated_at'. The main pane shows a SQL query editor with the following query:

```
1 DELETE FROM booking
2 WHERE ticket_price < 10000;
```

The query has been executed successfully, as indicated by the message in the Data Output pane: "DELETE 0. Query returned successfully in 42 msec." The status bar at the bottom shows "Total rows: Query complete 00:00:00.042".

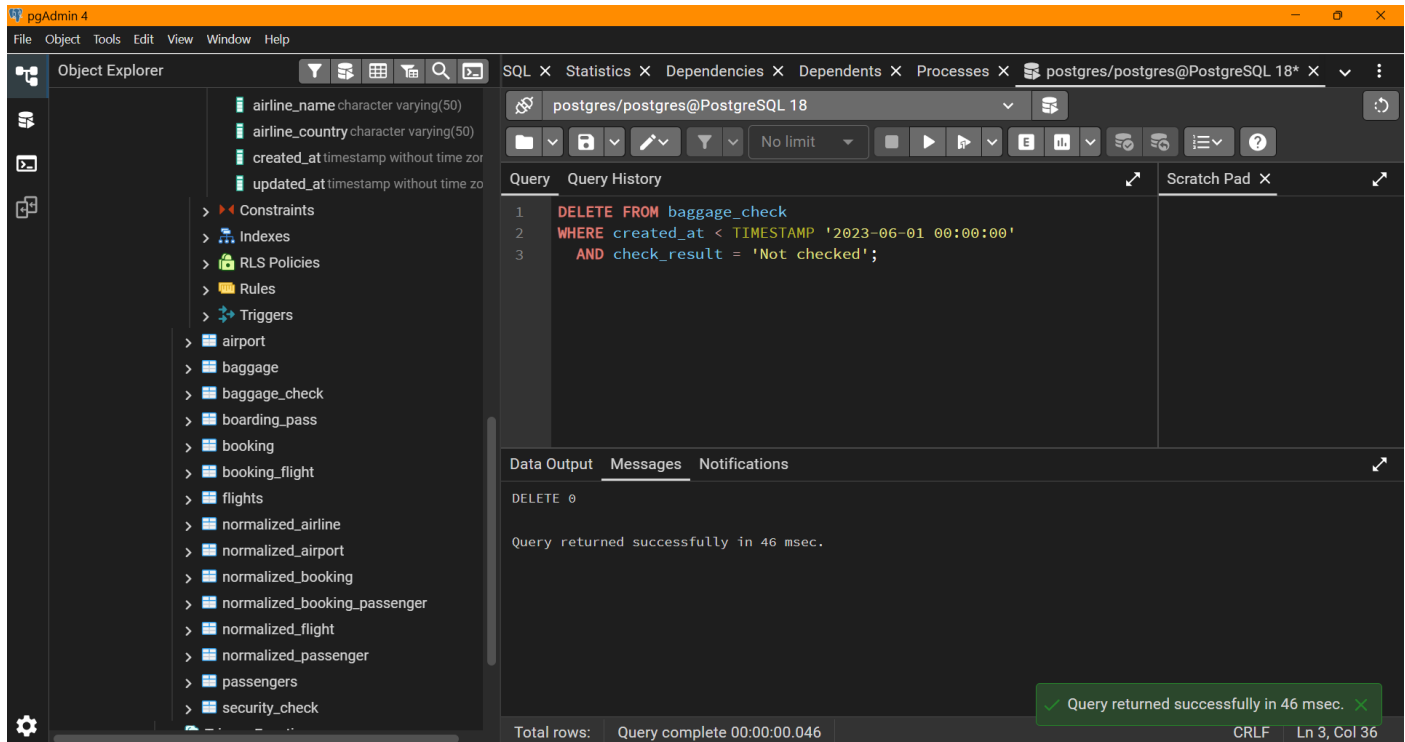
8. Set a default code 'UNK' (Unknown) for any airline in the airline table where airline_code is missing (NULL).

The screenshot shows the pgAdmin 4 interface. On the left, the Object Explorer displays a tree of database objects, including tables like 'airline_name', 'airline_country', 'created_at', and 'updated_at'. The main pane shows a SQL query editor with the following query:

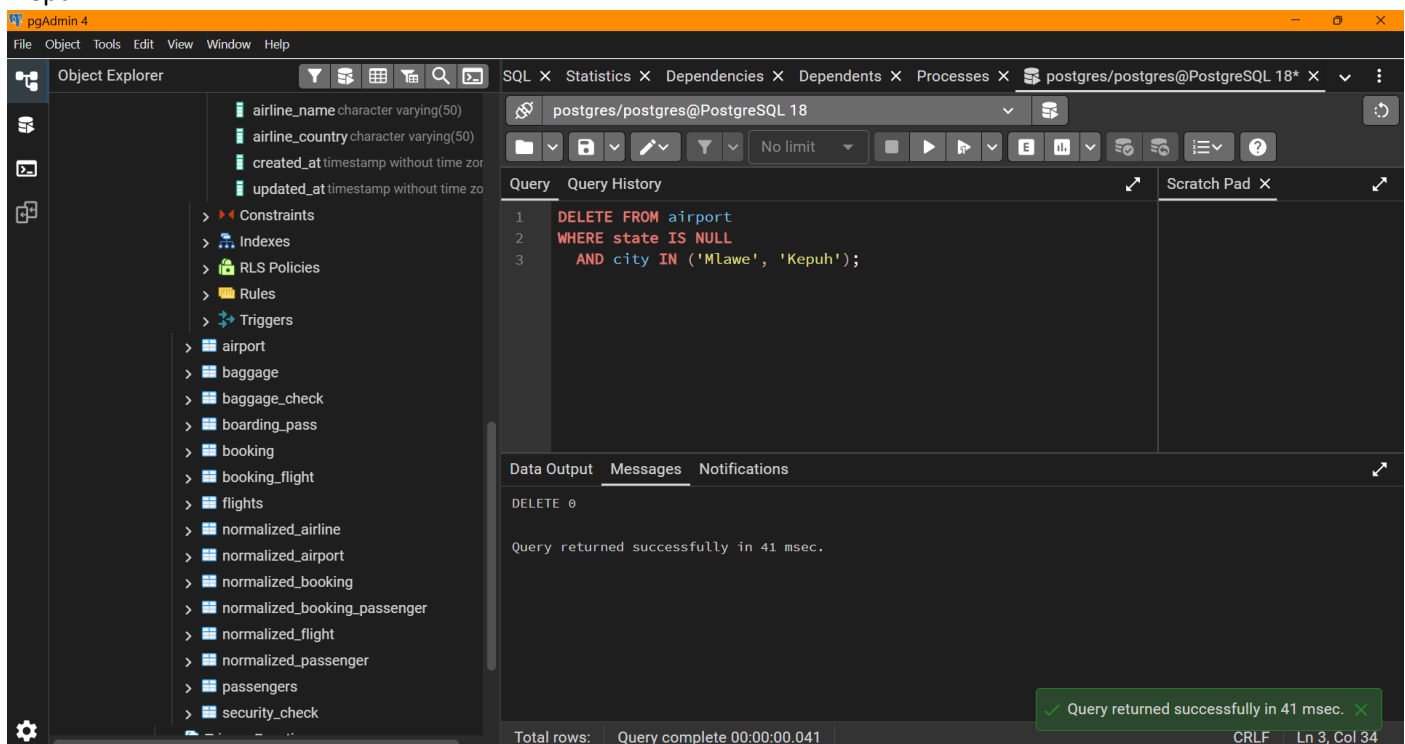
```
1 UPDATE airline
2 SET airline_code = 'UNK',
3     updated_at = CURRENT_TIMESTAMP
4 WHERE airline_code IS NULL;
```

The query has been executed successfully, as indicated by the message in the Data Output pane: "UPDATE 0. Query returned successfully in 54 msec." The status bar at the bottom shows "Total rows: Query complete 00:00:00.054".

9. Delete all baggage check records created before June 1, 2023, whose result is still 'Not checked'.



10. Delete all entries from the airport table where state is missing (NULL) and the city is either 'Mlawe' or 'Kepuh'.



11. Insert a new unchecked baggage inspection record and immediately return its newly generated baggage_check_id and created_at timestamp.

The screenshot shows the pgAdmin 4 interface. The Object Explorer on the left lists the database schema, including tables like 'airline', 'baggage_check', and 'normalized_booking_passenger'. The SQL query editor in the center contains the following SQL code:

```
INSERT INTO baggage_check (
    booking_id,
    passenger_id
)
VALUES (
    1,
    'Not checked',
    CURRENT_TIMESTAMP,
    CURRENT_TIMESTAMP,
    1,
    1
)
RETURNING baggage_check_id, created_at;
```

The Data Output pane at the bottom shows the result of the query:

baggage_check_id	created_at
1	2026-09-28 16:54:48.430544

Total rows: 1 Query complete 00:00:00.057

12. Convert all country names in the airline table to uppercase to ensure data consistency.

The screenshot shows the pgAdmin 4 interface. The Object Explorer on the left lists the database schema. The SQL query editor in the center contains the following SQL code:

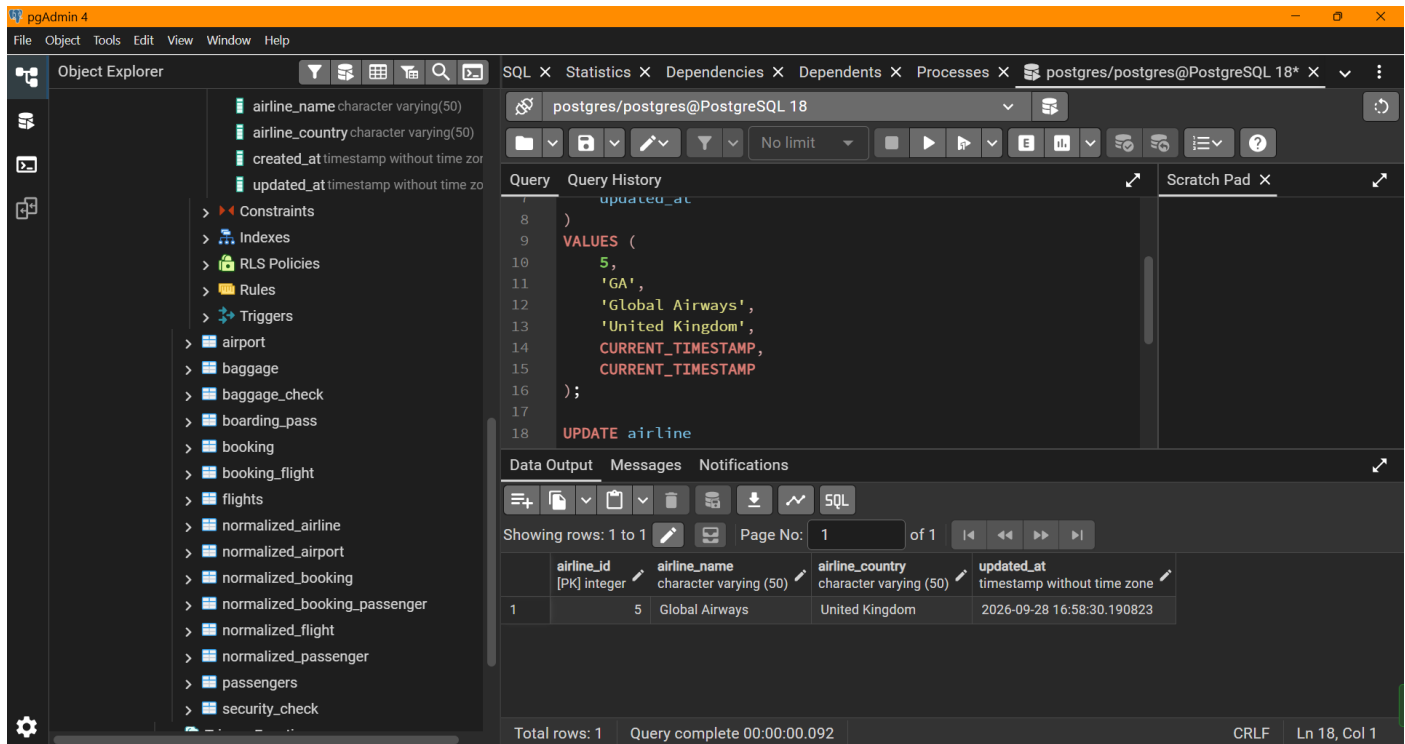
```
SELECT airline_id, airline_name, airline_country
FROM airline
ORDER BY airline_id;
```

The Data Output pane at the bottom shows the result of the query:

airline_id	airline_name	airline_country
1	KazAir	TURKEY
2	AirEasy	FRANCE
3	FlyFly	POLAND
4	FlyHigh	BRAZIL

Total rows: 4 Query complete 00:00:00.057

13. Update both the name and country for the airline with airline_id = 5. Set the name to 'Global Airways' and the country to 'United Kingdom', while updating update_at to the current date.



The screenshot shows the pgAdmin 4 interface. On the left, the Object Explorer displays the database schema, including tables like 'airline', 'airport', 'baggage', and 'normalized_*'. The central pane shows an SQL query being executed:

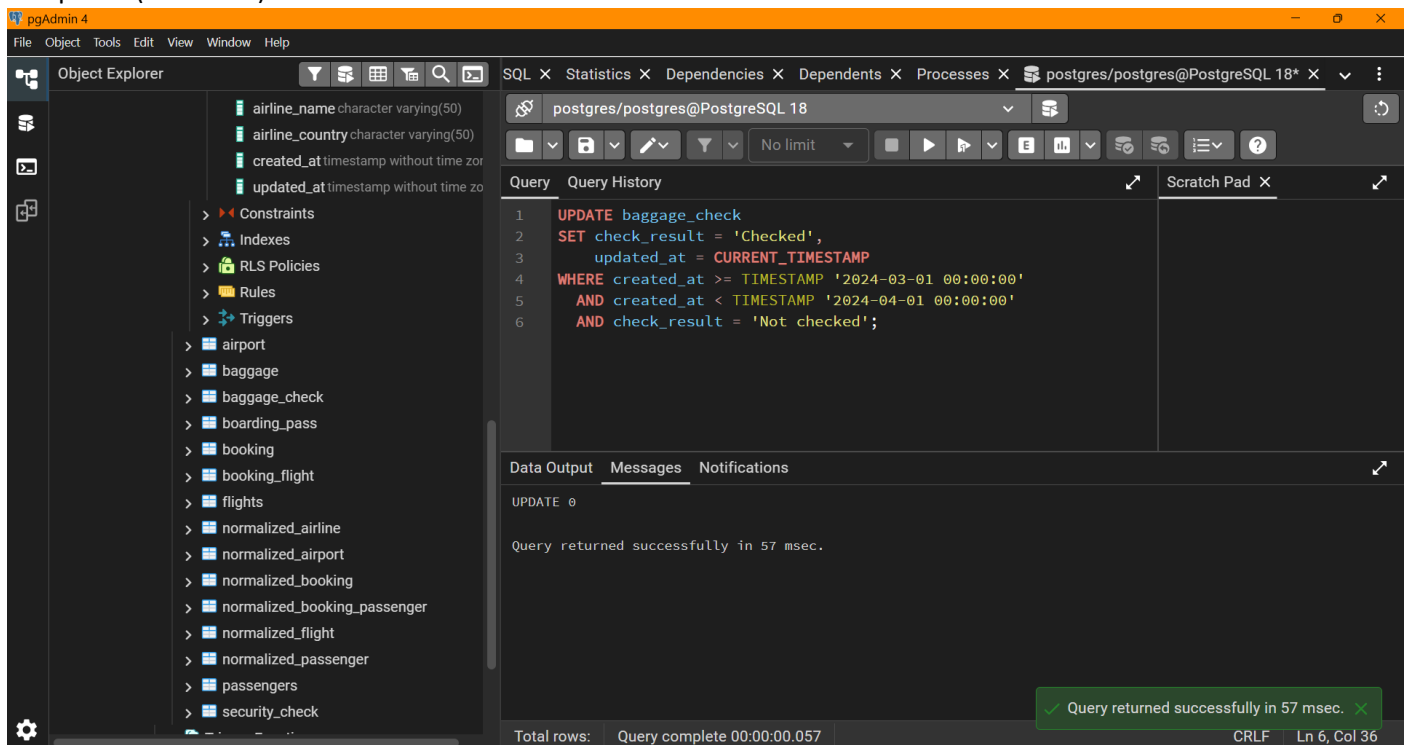
```
UPDATE airline
SET airline_name = 'Global Airways',
    airline_country = 'United Kingdom',
    updated_at = CURRENT_TIMESTAMP
WHERE airline_id = 5;
```

Below the query editor, the 'Data Output' tab shows the result of the update:

airline_id	airline_name	airline_country	updated_at
5	Global Airways	United Kingdom	2026-09-28 16:58:30.190823

The status bar at the bottom indicates 'Total rows: 1' and 'Query complete 00:00:00.092'.

14. Mark all baggage check records (baggage_check) created in March 2024 with a status of 'Not checked' as completed('Checked').



The screenshot shows the pgAdmin 4 interface. The central pane shows an SQL query being executed:

```
UPDATE baggage_check
SET check_result = 'Checked',
    updated_at = CURRENT_TIMESTAMP
WHERE created_at >= TIMESTAMP '2024-03-01 00:00:00'
AND created_at < TIMESTAMP '2024-04-01 00:00:00'
AND check_result = 'Not checked';
```

Below the query editor, the 'Data Output' tab shows the result of the update:

```
UPDATE 0
Query returned successfully in 57 msec.
```

A green notification box at the bottom right states: 'Query returned successfully in 57 msec.' The status bar at the bottom indicates 'Total rows: 0' and 'Query complete 00:00:00.057'.